ABSTRACT

CROP RECOMMENDATION AND PLANT DISEASE DETECTION SYSTEM USING MACHINE LEARNING AND CNN

Introduction :

Modern agriculture relies heavily on timely decision-making regarding crop selection and early pest management to maintain high yields and economic viability. However, traditional farming decisions are often based on intuition rather than data-driven insights, leading to suboptimal crop selection and unmitigated crop loss from disease.

Objective:

This project aims to develop a unified, user-friendly digital platform that empowers farmers by providing automated, highly accurate crop recommendations based on soil conditions and localized weather data, alongside an instant plant disease diagnostic tool using image processing.

Methodology:

The system features a secure user authentication portal connected to a responsive web interface. For crop prediction, multi-variable soil parameters (N, P, K, pH) combined with weather integration data are processed using a Supervised Machine Learning Classification model (such as Random Forest/XGBoost). For disease detection, users upload plant leaf images, which are processed by a Convolutional Neural Network (CNN) trained on thousands of plant leaf samples to identify specific pathogens and suggest targeted treatments.

Results:

The crop recommendation classifier achieved high predictive accuracy in matching soil-climate profiles to optimal crop types, while the CNN model demonstrated robust precision in identifying leaf diseases across multiple crop species, even under varying lighting conditions.

Conclusion

By combining soil analysis, environmental weather metrics, and computer vision into a single interface, this platform provides smallholder farmers with an accessible, real-time decision support tool that maximizes agricultural output, reduces disease-induced crop loss, and promotes sustainable farming practices.

Keywords:

Smart Agriculture, Machine Learning, Crop Recommendation, Plant Disease Detection, Convolutional Neural Networks (CNN), Weather Integration, Precision Farming.